

■ 시나리오 별 백업 복구(rman using recovery catalog)

KEY WORDS

- <시나리오 9> Undo DF 손상의 복구
- <시나리오 10> 정상 종료 후 CF 손상의 복구(백업본 사용)
- <시나리오 11> CF 손상의 복구(CF 재생성)
- <시나리오 12> 모든 DFs, CF 손상되었을 경우 복구
- <시나리오 13> CF 와 모든 RFS 손상의 복구
- <시나리오 14> 모든 DFs, CFs, RFS 와 모든 RFS 손상의 복구
- <시나리오 15> 시간을 기준으로 불완전 복구(Time based recovery)
- <시나리오 16> TS 에 속한 여러 DF 중 일부분이 손상된 경우 복구

<<시나리오 9>> Undo DF 손상의 복구

```
SYS@orcl> select salary from hr.employees where employee_id = 100;
SALARY
```

```
-----
30000
```

```
SYS@orcl> update hr.employees set salary = salary * 1.1 where employee_id = 100;
1 row updated.
```

```
SYS@orcl> select salary from hr.employees where employee_id = 100;
SALARY
```

```
-----
33000
```

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	ONLINE
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	ONLINE
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	ONLINE
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	ONLINE
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	ONLINE
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	ONLINE
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	ONLINE
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	ONLINE

9 _SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	ONLINE
10 _SYSSMU10_930580995\$	PUBLIC	UNDOTBS1	ONLINE

11 rows selected.

```
SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name,
2 t.ubafil undo_datafile_num, t.ubablk undo_block, t.used_ublk
3 from v$session s, v$transaction t, v$rollname r
4 where s.taddr = t.addr and t.xidusn = r.usn;
```

USERNAME

SID SERIAL# UNDO_SEG_NUM UNDO_SEG_NAME

UNDO_DATAFILE_NUM UNDO_BLOCK USED_UBLK

SYS

1	11279	10 _SYSSMU10_930580995\$
4	1725	1

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	4482740	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	4482740	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	4482740	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	4482740	ONLINE
5	SHTBS	/u01/app/oracle/oradata/ORCL/shtbs01.dbf	4482740	ONLINE
2	OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf	4482740	ONLINE
8	OETBS	/u01/app/oracle/oradata/ORCL/oetbs02.dbf	4483758	ONLINE

7 rows selected.

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/undotbs01.dbf: No such file or directory
```

```
SYS@orcl> alter system switch logfile;
System altered.
```

```
SYS@orcl> alter system checkpoint;
alter system checkpoint
*
ERROR at line 1:
ORA-03113: end-of-file on communication channel
Process ID: 11898
Session ID: 1 Serial number: 11279
```

```
SYS@orcl> conn /as sysdba
Connected to an idle instance.
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               822083584 bytes
Database Buffers   301989888 bytes
Redo Buffers              7880704 bytes
Database mounted.
ORA-01157: cannot identify/lock data file 4 - see DBWR trace file
ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
```

```
RMAN> select status from v$instance;
```

```
STATUS
-----
MOUNTED
```

```
RMAN> list failure;
```

```
Database Role: PRIMARY
```

```
List of Database Failures
```

```
=====
```

Failure ID	Priority	Status	Time Detected	Summary
2	HIGH	OPEN	19-MAY-25	One or more non-system datafiles are missing

RMAN> list failure 2 detail;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
2	HIGH	OPEN	19-MAY-25	One or more non-system datafiles are missing

Impact: See impact for individual child failures

List of child failures for parent failure ID 2

Failure ID	Priority	Status	Time Detected	Summary
6845	HIGH	OPEN	19-MAY-25	Datafile 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

6845 HIGH OPEN 19-MAY-25 Datafile 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

is missing

Impact: Some objects in tablespace UNDOTBS1 might be unavailable

RMAN> restore tablespace undotbs1;

Starting restore at 19-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=46 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00004 to /u01/app/oracle/oradata/ORCL/undotbs01.dbf

channel	ORA_DISK_1:	reading	from	backup	piece

channel	ORA_DISK_1:	piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T021707_n2kk4mp1_.

bkp tag=TAG20250518T021707

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:01

Finished restore at 19-MAY-25

RMAN> recover tablespace undotbs1;

Starting recover at 19-MAY-25

using channel ORA_DISK_1

starting media recovery

```

/home/oracle/arch1/arch_1_40_1201218512.arc
archived log for thread 1 with sequence 41 is already on disk as file
/home/oracle/arch1/arch_1_41_1201218512.arc
archived log for thread 1 with sequence 42 is already on disk as file
/home/oracle/arch1/arch_1_42_1201218512.arc

```

```

-----
archived log file name=/home/oracle/arch1/arch_1_38_1201218512.arc thread=1 sequence=38
archived log file name=/home/oracle/arch1/arch_1_39_1201218512.arc thread=1 sequence=39
archived log file name=/home/oracle/arch1/arch_1_40_1201218512.arc thread=1 sequence=40
media recovery complete, elapsed time: 00:00:01
Finished recover at 19-MAY-25

```

```

RMAN> alter database open;
Statement processed

```

```

RMAN> select salary from hr.employees where employee_id = 100;
      SALARY
-----
      30000

```

- Undo 확인

```

SYS@orcl> update hr.employees set salary = salary * 1.1 where employee_id = 100;
1 row updated.

```

```

SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name, t.ubafil
undo_datafile_num, t.ubablk undo_block, t.used_ublk from v$session s, v$transaction t, v$rollname r where s.taddr
= t.addr and t.xidusn = r.usn;
USERNAME

```

```

-----
      SID      SERIAL# UNDO_SEG_NUM UNDO_SEG_NAME
-----
UNDO_DATAFILE_NUM UNDO_BLOCK  USED_UBLK
-----
SYS
      1         50353          1 _SYSSMU1_1261223759$
      4          503           1

```

```

SYS@orcl> rollback;
Rollback complete.

```

```

RMAN> list backup;

```

<<시나리오 10>> 정상 종료 후 CF 손상의 복구

- 정상 종료

```
SYS@orcl> shut immediate
```

```
SYS@orcl> !ls /u01/app/oracle/oradata/ORCL
```

```
control01.ctl oetbs01.dbf redo02.log sysaux01.dbf undotbs01.dbf
control02.ctl oetbs02.dbf redo03.log system01.dbf users01.dbf
initorcl.ora redo01.log shtbs01.dbf temp01.dbf
```

- CF, autobackup CF확인

```
[oracle@oel7v9r1 2025_05_19]$ pwd
```

```
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19
```

```
[oracle@oel7v9r1 2025_05_19]$ ll
```

```
total 21504
```

```
-rw-r-----. 1 oracle dba 11010048 May 19 00:02 o1_mf_s_1201478527_n2mxmhqh_.bkp
```

```
-rw-r-----. 1 oracle dba 11010048 May 19 00:22 o1_mf_s_1201479776_n2mytjol_.bkp
```

```
RMAN> LIST BACKUP OF CONTROLFILE ;
```

```
List of Backup Sets
```

```
=====
```

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

```
-----
```

1052	Full		10.30M	DISK		00:00:00	18-MAY-25
------	------	--	--------	------	--	----------	-----------

```
BP Key: 1058 Status: AVAILABLE Compressed: NO Tag: TAG20250518T021716
```

```
Piece
```

Name:

```
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201400236_n2kk4w2k_.bkp
```

```
Control File Included: Ckp SCN: 3555033 Ckp time: 18-MAY-25
```

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

```
-----
```

1409	Full		10.30M	DISK		00:00:00	18-MAY-25
------	------	--	--------	------	--	----------	-----------

```
BP Key: 1412 Status: AVAILABLE Compressed: NO Tag: TAG20250518T214910
```

```
Piece
```

Name:

```
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201470550_n2mot6wj_.bkp
```

```
Control File Included: Ckp SCN: 4379296 Ckp time: 18-MAY-25
```

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

```
-----
```

1452	Full		10.48M	DISK		00:00:00	18-MAY-25
------	------	--	--------	------	--	----------	-----------

```
BP Key: 1454 Status: AVAILABLE Compressed: NO Tag: TAG20250518T234222
```

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201477342_n2mwggnh_.bkp

Control File Included: Ckp SCN: 4381345 Ckp time: 18-MAY-25

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time
1499	Full	10.48M	DISK	00:00:00	18-MAY-25
BP Key: 1502 Status: AVAILABLE Compressed: NO Tag: TAG20250518T235357					

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201478037_n2mx45gn_.bkp

Control File Included: Ckp SCN: 4381984 Ckp time: 18-MAY-25

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time
1554	Full	10.48M	DISK	00:00:00	19-MAY-25
BP Key: 1556 Status: AVAILABLE Compressed: NO Tag: TAG20250519T000207					

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201478527_n2mxmhqh_.bkp

Control File Included: Ckp SCN: 4382504 Ckp time: 19-MAY-25

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time
1613	Full	10.48M	DISK	00:00:00	19-MAY-25
BP Key: 1616 Status: AVAILABLE Compressed: NO Tag: TAG20250519T002256					

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201479776_n2mytjol_.bkp

Control File Included: Ckp SCN: 4483949 Ckp time: 19-MAY-25

- 장애 발생

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/control01.ctl

SYS@orcl> conn hr/hr

ERROR:

ORA-01034: ORACLE not available

ORA-27101: shared memory realm does not exist

Linux-x86_64 Error: 2: No such file or directory

Additional information: 4376

Additional information: -1207692525

Process ID: 0

Session ID: 0 Serial number: 0

```
[oracle@oel7v9r1 ~]$ sqlplus / as sysdba
SYS@orcl> startup nomount
```

```
[oracle@oel7v9r1 ~]$ rman target "sys/oracle@t1 as sysdba" catalog ruser/ruser@c1
connected to target database: ORCL (not mounted)
connected to recovery catalog database
```

- Recovery

```
RMAN> restore controlfile from
'/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201479776_n2mytjol_.bkp'; <-- 최근 백업본
Starting restore at 19-MAY-25
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=39 device type=DISK

channel ORA_DISK_1: restoring control file
channel ORA_DISK_1: restore complete, elapsed time: 00:00:01
output file name=/u01/app/oracle/oradata/ORCL/control01.ctl
Finished restore at 19-MAY-25
```

```
RMAN> alter database mount;
released channel: ORA_DISK_1
Statement processed
```

```
RMAN> recover database; <-- DF손상이 없으니....
Starting recover at 19-MAY-25
Starting implicit crosscheck backup at 19-MAY-25
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=46 device type=DISK
Crosschecked 19 objects
Finished implicit crosscheck backup at 19-MAY-25
```

```
Starting implicit crosscheck copy at 19-MAY-25
using channel ORA_DISK_1
Finished implicit crosscheck copy at 19-MAY-25
```

```
searching for all files in the recovery area
cataloging files...
cataloging done
```

List of Cataloged Files

=====

File Name: /home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201479776_n2mytjol_.bkp
using channel ORA_DISK_1
starting media recovery

archived log for thread 1 with sequence 42 is already on disk as file
/home/oracle/arch1/arch_1_42_1201218512.arc
archived log for thread 1 with sequence 43 is already on disk as file
/home/oracle/arch1/arch_1_43_1201218512.arc
archived log for thread 1 with sequence 46 is already on disk as file /u01/app/oracle/oradata/ORCL/redo01.log
archived log for thread 1 with sequence 47 is already on disk as file /u01/app/oracle/oradata/ORCL/redo02.log
archived log for thread 1 with sequence 48 is already on disk as file /u01/app/oracle/oradata/ORCL/redo03.log
archived log file name=/home/oracle/arch1/arch_1_42_1201218512.arc thread=1 sequence=42
archived log file name=/home/oracle/arch1/arch_1_43_1201218512.arc thread=1 sequence=43
archived log file name=/home/oracle/arch1/arch_1_44_1201218512.arc thread=1 sequence=44
archived log file name=/home/oracle/arch1/arch_1_45_1201218512.arc thread=1 sequence=45
archived log file name=/u01/app/oracle/oradata/ORCL/redo01.log thread=1 sequence=46
archived log file name=/u01/app/oracle/oradata/ORCL/redo02.log thread=1 sequence=47
archived log file name=/u01/app/oracle/oradata/ORCL/redo03.log thread=1 sequence=48
media recovery complete, elapsed time: 00:00:00
Finished recover at 19-MAY-25

RMAN> alter database open resetlogs;
Statement processed
new incarnation of database registered in recovery catalog
starting full resync of recovery catalog
full resync complete

- Tag사용하여 Incremental Level 0 Backup 하기

[oracle@oel7v9r1 ~]\$ vi bkp_full.rcv

run

```
{  
    backup as compressed backupset incremental level 0 database tag '&1';  
}
```

[oracle@oel7v9r1 ~]\$ export MYTAG='BKP_FULL_'\$(date +%Y%m%d%H%M%S)

[oracle@oel7v9r1 ~]\$ echo \$MYTAG

BKP_FULL_20250519223158

[oracle@oel7v9r1 ~]\$ rman target / cmdfile=bpk_full.rcv using \$MYTAG catalog ruser/ruser@c1

connected to target database: ORCL (DBID=1727714120)

connected to recovery catalog database

```

RMAN> run{
2> backup as compressed backupset incremental level 0 database tag 'BKP_FULL_20250519223158';
3> }
4>
Starting backup at 19-MAY-25
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=83 device type=DISK
channel ORA_DISK_1: starting compressed incremental level 0 datafile backup set
channel ORA_DISK_1: specifying datafile(s) in backup set
input datafile file number=00001 name=/u01/app/oracle/oradata/ORCL/system01.dbf
input datafile file number=00003 name=/u01/app/oracle/oradata/ORCL/sysaux01.dbf
input datafile file number=00004 name=/u01/app/oracle/oradata/ORCL/undotbs01.dbf
input datafile file number=00002 name=/u01/app/oracle/oradata/ORCL/oetbs01.dbf
input datafile file number=00005 name=/u01/app/oracle/oradata/ORCL/shtbs01.dbf
input datafile file number=00008 name=/u01/app/oracle/oradata/ORCL/oetbs02.dbf
input datafile file number=00007 name=/u01/app/oracle/oradata/ORCL/users01.dbf
channel ORA_DISK_1: starting piece 1 at 19-MAY-25
channel ORA_DISK_1: finished piece 1 at 19-MAY-25
piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_19/o1_mf_nnnd0_BKP_FULL_20250519223_n2pdx2cd_
.bkp tag=BKP_FULL_20250519223158 comment=NONE
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:35
Finished backup at 19-MAY-25

Starting Control File and SPFILE Autobackup at 19-MAY-25
piece      handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201559766_n2pdy6fo_.bkp
comment=NONE
Finished Control File and SPFILE Autobackup at 19-MAY-25

Recovery Manager complete.

```

```
RMAN> list backup tag 'BKP_FULL_20250519223158';
```

```
List of Backup Sets
```

```
=====
```

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

```
-----
```

1997	Incr 0	299.55M	DISK		00:00:29	19-MAY-25
------	--------	---------	------	--	----------	-----------

BP Key: 2000	Status: AVAILABLE	Compressed: YES	Tag: BKP_FULL_20250519223158
--------------	-------------------	-----------------	------------------------------

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_19/o1_mf_nnnd0_BKP_FULL_20250519223_n2pdx2cd_.bkp

List of Datafiles in backup set 1997

File LV Type Ckp SCN Ckp Time Abs Fuz SCN Sparse Name

1	0	Incr	4706543	19-MAY-25		NO	/u01/app/oracle/oradata/ORCL/system01.dbf
2	0	Incr	4706543	19-MAY-25		NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	0	Incr	4706543	19-MAY-25		NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	0	Incr	4706543	19-MAY-25		NO	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	0	Incr	4706543	19-MAY-25		NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	0	Incr	4706543	19-MAY-25		NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	0	Incr	4706543	19-MAY-25		NO	/u01/app/oracle/oradata/ORCL/oetbs02.dbf

RMAN> report obsolete;

RMAN> delete obsolete;

RMAN> list expired archivelog all;

RMAN> delete expired archivelog all;

RMAN> alter system archive log current;

RMAN> list archivelog all;

<<시나리오 11>> CF 손상의 복구(CF 재생성)

```
RMAN> select name from v$controlfile;
```

```
NAME
```

```
-----  
/u01/app/oracle/oradata/ORCL/control01.ctl
```

```
RMAN> host 'rm /u01/app/oracle/oradata/ORCL/control01.ctl';
```

```
host command complete
```

```
RMAN> host 'ls /u01/app/oracle/oradata/ORCL/control01.ctl';
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/control01.ctl: No such file or directory
```

```
host command complete
```

```
RMAN-00571: =====
```

```
RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====
```

```
RMAN-00571: =====
```

```
RMAN-06135: error executing host command: Additional information: 512
```

```
RMAN> shutdown abort
```

```
Oracle instance shut down
```

```
RMAN> startup nomount
```

```
connected to target database (not started)
```

```
Oracle instance started
```

```
Total System Global Area      1140849904 bytes
```

```
Fixed Size                      8895728 bytes
```

```
Variable Size                   822083584 bytes
```

```
Database Buffers                301989888 bytes
```

```
Redo Buffers                     7880704 bytes
```

```
-- Recovery
```

```
RMAN> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
STARTED
```

```
RMAN> restore controlfile from autobackup;
```

```
Starting restore at 19-MAY-25
```

```
allocated channel: ORA_DISK_1
```

```
channel ORA_DISK_1: SID=38 device type=DISK
recovery area destination: /home/oracle/backup/FRA
database name (or database unique name) used for search: ORCL
channel ORA_DISK_1: AUTOBACKUP
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201559766_n2pdy6fo_.bkp found in the
recovery area
channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250519
channel ORA_DISK_1: restoring control file from AUTOBACKUP
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201559766_n2pdy6fo_.bkp
channel ORA_DISK_1: control file restore from AUTOBACKUP complete
output file name=/u01/app/oracle/oradata/ORCL/control01.ctl
Finished restore at 19-MAY-25
```

```
RMAN> alter database mount;
released channel: ORA_DISK_1
Statement processed
```

```
RMAN> alter database backup controlfile to trace as '/home/oracle/backup/crecon_20250520.txt';
Statement processed
RMAN> shutdown abort
Oracle instance shut down
```

```
[oracle@oel7v9r1 backup]$ vi /home/oracle/backup/crecon_20250520.txt
STARTUP NOMOUNT
CREATE CONTROLFILE REUSE DATABASE "ORCL" NORESETLOGS ARCHIVELOG
    MAXLOGFILES 16
    MAXLOGMEMBERS 3
    MAXDATAFILES 100
    MAXINSTANCES 8
    MAXLOGHISTORY 292
LOGFILE
    GROUP 1 '/u01/app/oracle/oradata/ORCL/redo01.log' SIZE 10M BLOCKSIZE 512,
    GROUP 2 '/u01/app/oracle/oradata/ORCL/redo02.log' SIZE 10M BLOCKSIZE 512,
    GROUP 3 '/u01/app/oracle/oradata/ORCL/redo03.log' SIZE 10M BLOCKSIZE 512
-- STANDBY LOGFILE
DATAFILE
    '/u01/app/oracle/oradata/ORCL/system01.dbf',
    '/u01/app/oracle/oradata/ORCL/oetbs01.dbf',
    '/u01/app/oracle/oradata/ORCL/sysaux01.dbf',
    '/u01/app/oracle/oradata/ORCL/undotbs01.dbf',
    '/u01/app/oracle/oradata/ORCL/shtbs01.dbf',
```

```
'/u01/app/oracle/oradata/ORCL/users01.dbf',  
'/u01/app/oracle/oradata/ORCL/oetbs02.dbf'  
CHARACTER SET AL32UTF8  
;  
RECOVER DATABASE  
ALTER SYSTEM ARCHIVE LOG ALL;  
ALTER DATABASE OPEN;  
ALTER TABLESPACE TEMP ADD TEMPFILE '/u01/app/oracle/oradata/ORCL/temp01.dbf' REUSE;
```

```
SYS@orcl> @/home/oracle/backup/crecon_20250520.txt  
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes  
Fixed Size                  8895728 bytes  
Variable Size               822083584 bytes  
Database Buffers   301989888 bytes  
Redo Buffers          7880704 bytes
```

```
Control file created.  
Media recovery complete.  
System altered.  
Database altered.  
Tablespace altered.
```

- 백업하기

```
[oracle@oel7v9r1 ~]$ rman target "sys/oracle@t1 as sysdba" catalog ruser/ruser@c1  
connected to target database: ORCL (DBID=1727714120)  
connected to recovery catalog database
```

```
RMAN> CREATE SCRIPT full_backup  
{  
  BACKUP DATABASE PLUS ARCHIVELOG;  
  DELETE OBSOLETE;  
}
```

```
starting full resync of recovery catalog  
full resync complete  
created script full_backup
```

```
RMAN> RUN { EXECUTE SCRIPT full_backup; }  
executing script: full_backup
```

RMAN> PRINT SCRIPT full_backup;

printing stored script: full_backup

```
{  
  BACKUP DATABASE PLUS ARCHIVELOG;  
  DELETE OBSOLETE;  
}
```

RMAN> LIST ALL SCRIPT NAMES;

List of Stored Scripts in Recovery Catalog

Scripts of Target Database ORCL

Script Name

Description

full_backup

- 셸에서 스크립터 수행해서 백업하기

[oracle@oel7v9r1 ~]\$ rman target "sys/oracle@t1 as sysdba" catalog ruser/ruser@c1 script full_backup;

connected to target database: ORCL (DBID=1727714120)

connected to recovery catalog database

executing script: full_backup

- log miner 사용 모드.

SYS@orcl> select supplemental_log_data_min from v\$database;

SUPPLEME

NO

SYS@orcl> alter database add supplemental log data;

Database altered.

SYS@orcl> select supplemental_log_data_min from v\$database;

SUPPLEME

YES

<<시나리오 12>> 모든 DFs, CF 손상의 복구

```
SYS@orcl> alter system switch logfile; <--- 여러번 수행
System altered.
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.{dbf,ctl}
SYS@orcl> alter system checkpoint;
System altered.
```

```
SYS@orcl> conn hr/hr
ERROR:
ORA-00604: error occurred at recursive SQL level 1
ORA-01116: error in opening database file 1
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
```

```
RMAN> restore controlfile from autobackup;
```

```
Starting restore at 20-MAY-25
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=39 device type=DISK
```

```
recovery area destination: /home/oracle/backup/FRA
database name (or database unique name) used for search: ORCL
channel ORA_DISK_1: AUTOBACKUP
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201563948_n2pk0w8z_bkp found in the
recovery area
channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250520
channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250519
channel ORA_DISK_1: restoring control file from AUTOBACKUP
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201563948_n2pk0w8z_bkp
channel ORA_DISK_1: control file restore from AUTOBACKUP complete
output file name=/u01/app/oracle/oradata/ORCL/control01.ctl
Finished restore at 20-MAY-25
```

```
RMAN> alter database mount;
```

```
released channel: ORA_DISK_1
Statement processed
```


RMAN> **restore database;**

Starting restore at 20-MAY-25

Starting implicit crosscheck backup at 20-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=40 device type=DISK

Crosschecked 6 objects

Finished implicit crosscheck backup at 20-MAY-25

Starting implicit crosscheck copy at 20-MAY-25

using channel ORA_DISK_1

Finished implicit crosscheck copy at 20-MAY-25

searching for all files in the recovery area

cataloging files...

cataloging done

List of Cataloged Files

=====

File Name: /home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201563948_n2pk0w8z_bkp

using channel ORA_DISK_1

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/ORCL/system01.dbf

channel ORA_DISK_1: restoring datafile 00002 to /u01/app/oracle/oradata/ORCL/oetbs01.dbf

channel ORA_DISK_1: restoring datafile 00003 to /u01/app/oracle/oradata/ORCL/sysaux01.dbf

channel ORA_DISK_1: restoring datafile 00004 to /u01/app/oracle/oradata/ORCL/undotbs01.dbf

channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL/shtbs01.dbf

channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: restoring datafile 00008 to /u01/app/oracle/oradata/ORCL/oetbs02.dbf

channel ORA_DISK_1: reading from backup piece

/home/oracle/backup/FRA/ORCL/backupset/2025_05_19/o1_mf_nnndf_TAG20250519T234537_n2pk0lkw_bkp

channel ORA_DISK_1: piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_19/o1_mf_nnndf_TAG20250519T234537_n2pk0lkw_b

kp tag=TAG20250519T234537

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:07

Finished restore at 20-MAY-25

RMAN> recover database;

Starting recover at 20-MAY-25

using channel ORA_DISK_1

starting media recovery

archived log for thread 1 with sequence 6 is already on disk as file /home/oracle/arch1/arch_1_6_1201558447.arc

archived log for thread 1 with sequence 9 is already on disk as file /u01/app/oracle/oradata/ORCL/redo03.log

archived log for thread 1 with sequence 10 is already on disk as file /u01/app/oracle/oradata/ORCL/redo01.log

archived log for thread 1 with sequence 11 is already on disk as file /u01/app/oracle/oradata/ORCL/redo02.log

archived log file name=/home/oracle/arch1/arch_1_6_1201558447.arc thread=1 sequence=6

archived log file name=/home/oracle/arch1/arch_1_7_1201558447.arc thread=1 sequence=7

archived log file name=/home/oracle/arch1/arch_1_8_1201558447.arc thread=1 sequence=8

archived log file name=/u01/app/oracle/oradata/ORCL/redo03.log thread=1 sequence=9

archived log file name=/u01/app/oracle/oradata/ORCL/redo01.log thread=1 sequence=10

archived log file name=/u01/app/oracle/oradata/ORCL/redo02.log thread=1 sequence=11

media recovery complete, elapsed time: 00:00:00

Finished recover at 20-MAY-25

RMAN> alter database open resetlogs;

Statement processed

new incarnation of database registered in recovery catalog

starting full resync of recovery catalog

full resync complete

- 백업받기

RMAN> replace script full_backup

2> {

3> BACKUP AS BACKUPSET DATABASE PLUS ARCHIVELOG;

4> crosscheck archivelog all;

5> delete obsolete;

6> delete expired archivelog all;

7> }

replaced script full_backup

RMAN> RUN { EXECUTE SCRIPT full_backup; }

<<시나리오 13>> CF과 모든 RFs 손상의 복구

RMAN> list archivelog all;

List of Archived Log Copies for database with db_unique_name ORCL

```
=====
Key      Thrd Seq      S Low Time
-----
2505     1    4        A 20-MAY-25
          Name: /home/oracle/arch1/arch_1_4_1201565110.arc
```

RMAN> select * from v\$database_incarnation;

INCARNATION# RESETLOGS_CHANGE# RESETLOGS PRIOR_RESETLOGS_CHANGE# PRIOR_RES

```
-----
STATUS  RESETLOGS_ID PRIOR_INCARNATION# FLASHBACK_DATABASE_ALLOWED  CON_ID
-----
          1          4704513 19-MAY-25          3114462 15-MAY-25
PARENT  1201558447          0 NO          0
          2          4913726 20-MAY-25          4704513 19-MAY-25
CURRENT 1201565110          1 NO          0
```

- 장애 발생

SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,log}

/u01/app/oracle/oradata/ORCL/control01.ctl

/u01/app/oracle/oradata/ORCL/redo01.log

/u01/app/oracle/oradata/ORCL/redo02.log

/u01/app/oracle/oradata/ORCL/redo03.log

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,log}

SYS@orcl> alter system checkpoint;

System altered.

[oracle@oel7v9r1 ~]\$ rman target "sys/oracle@t1 as sysdba" catalog ruser/ruser@c1

RMAN-00571: =====

RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====

RMAN-00571: =====

RMAN-00554: initialization of internal recovery manager package failed

RMAN-06003: Oracle error from target database:

ORA-00210: cannot open the specified control file

ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'

ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3

SYS@orcl> shut abort

SYS@orcl> shut abort
ORACLE instance shut down.
SYS@orcl> startup nomount
ORACLE instance started.

Total System Global Area 1140849904 bytes
Fixed Size 8895728 bytes
Variable Size 822083584 bytes
Database Buffers 301989888 bytes
Redo Buffers 7880704 bytes

- Recovery

RMAN> restore controlfile from autobackup;

RMAN> alter database mount;

RMAN> restore database;

RMAN> recover database;

Starting recover at 20-MAY-25
using channel ORA_DISK_1

starting media recovery

archived log for thread 1 with sequence 4 is already on disk as file /home/oracle/arch1/arch_1_4_1201565110.arc
archived log file name=/home/oracle/arch1/arch_1_4_1201565110.arc thread=1 sequence=4

unable to find archived log

archived log thread=1 sequence=5

RMAN-00571: =====

RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====

RMAN-00571: =====

RMAN-03002: failure of recover command at 05/20/2025 00:27:51

RMAN-06054: media recovery requesting unknown archived log for thread 1 with sequence 5 and starting SCN
of **4915958**

```
RMAN> run {
```

```
2> set until scn=4915958;
```

```
3> restore database;
```

```
4> recover database;
```

```
5> }
```

executing command: SET until clause

Starting restore at 20-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/ORCL/system01.dbf

channel ORA_DISK_1: restoring datafile 00002 to /u01/app/oracle/oradata/ORCL/oetbs01.dbf

channel ORA_DISK_1: restoring datafile 00003 to /u01/app/oracle/oradata/ORCL/sysaux01.dbf

channel ORA_DISK_1: restoring datafile 00004 to /u01/app/oracle/oradata/ORCL/undotbs01.dbf

channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL/shtbs01.dbf

channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: restoring datafile 00008 to /u01/app/oracle/oradata/ORCL/oetbs02.dbf

channel ORA_DISK_1: reading from backup piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T001426_n2plpm67_.bkp

channel ORA_DISK_1: piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T001426_n2plpm67_.bkp tag=TAG20250520T001426

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:07

Finished restore at 20-MAY-25

Starting recover at 20-MAY-25

using channel ORA_DISK_1

starting media recovery

archived log for thread 1 with sequence 4 is already on disk as file /home/oracle/arch1/arch_1_4_1201565110.arc

archived log file name=/home/oracle/arch1/arch_1_4_1201565110.arc thread=1 sequence=4

media recovery complete, elapsed time: 00:00:00

Finished recover at 20-MAY-25

```
RMAN> alter database open resetlogs;
```

```
RMAN> RUN { EXECUTE SCRIPT full_backup; }
```

```

RMAN> print SCRIPT full_backup;
printing stored script: full_backup
{
BACKUP AS BACKUPSET DATABASE PLUS ARCHIVELOG;
crosscheck archivelog all;
delete obsolete;
delete expired archivelog all;
}

```

```

RMAN> replace SCRIPT full_backup
2> { BACKUP AS BACKUPSET DATABASE PLUS ARCHIVELOG;
3> crosscheck archivelog all;
4> delete obsolete;
5> delete expired archivelog all;
6> alter system archive log current;
7> }

```

replaced script full_backup

```

RMAN> list archivelog all;

```

List of Archived Log Copies for database with db_unique_name ORCL

=====

Key	Thrd	Seq	S	Low	Time
2618	1	2	A		20-MAY-25
Name: /home/oracle/arch1/arch_1_2_1201566881.arc					

```

RMAN> host 'ls /home/oracle/arch1';
arch_1_2_1201566881.arc
host command complete

```

<<시나리오 14>> 모든 DFs, RFs, CFs과 모든 AFs 손상의 복구

- 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/
```

```
SYS@orcl> ! rm /home/oracle/arch1/*
```

```
SYS@orcl> conn hr/hr
```

ERROR:

2025-05-20T00:46:55.064963+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_26895.trc:

ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'

ORA-27041: unable to open file

Linux-x86_64 Error: 2: No such file or directory

```
SYS@orcl> shut abort
```

ORACLE instance shut down.

- Recovery

```
RMAN> startup nomount
```

Oracle instance started

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 822083584 bytes

Database Buffers 301989888 bytes

Redo Buffers 7880704 bytes

```
RMAN> restore controlfile from autobackup;
```

Starting restore at 20-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=38 device type=DISK

recovery area destination: /home/oracle/backup/FRA

database name (or database unique name) used for search: ORCL

```
channel                                ORA_DISK_1:                                AUTOBACKUP
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201566934_n2pmy6op_.bkp found in the
recovery area
channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250520
channel      ORA_DISK_1:      restoring      control      file      from      AUTOBACKUP
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201566934_n2pmy6op_.bkp
channel ORA_DISK_1: control file restore from AUTOBACKUP complete
output file name=/u01/app/oracle/oradata/ORCL/control01.ctl
Finished restore at 20-MAY-25
```

```
RMAN> alter database mount;
```

```
released channel: ORA_DISK_1
Statement processed
```

```
RMAN> restore database;
```

```
Starting restore at 20-MAY-25
Starting implicit crosscheck backup at 20-MAY-25
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=38 device type=DISK
Crosschecked 8 objects
Finished implicit crosscheck backup at 20-MAY-25
```

```
Starting implicit crosscheck copy at 20-MAY-25
using channel ORA_DISK_1
Finished implicit crosscheck copy at 20-MAY-25
```

```
searching for all files in the recovery area
cataloging files...
cataloging done
```

```
List of Cataloged Files
```

```
=====
```

```
File Name: /home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201566934_n2pmy6op_.bkp
```

```
using channel ORA_DISK_1
```

```
channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/ORCL/system01.dbf
```



```

channel ORA_DISK_1: restoring datafile 00002 to /u01/app/oracle/oradata/ORCL/oetbs01.dbf
channel ORA_DISK_1: restoring datafile 00003 to /u01/app/oracle/oradata/ORCL/sysaux01.dbf
channel ORA_DISK_1: restoring datafile 00004 to /u01/app/oracle/oradata/ORCL/undotbs01.dbf
channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL/shtbs01.dbf
channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf
channel ORA_DISK_1: restoring datafile 00008 to /u01/app/oracle/oradata/ORCL/oetbs02.dbf
channel          ORA_DISK_1:          reading          from          backup          piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T003527_n2pmy0wh_.bkp
channel          ORA_DISK_1:          piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T003527_n2pmy0wh_.
bkp tag=TAG20250520T003527
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:00:07
Finished restore at 20-MAY-25

```

RMAN> recover database;

Starting recover at 20-MAY-25
using channel ORA_DISK_1

starting media recovery

```

channel ORA_DISK_1: starting archived log restore to default destination
channel ORA_DISK_1: restoring archived log
archived log thread=1 sequence=2
channel          ORA_DISK_1:          reading          from          backup          piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_anndf_TAG20250520T003532_n2pmy4td_.bkp
channel          ORA_DISK_1:          piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_anndf_TAG20250520T003532_n2pmy4td_.
bkp tag=TAG20250520T003532
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:00:01
archived log file name=/home/oracle/arch1/arch_1_2_1201566881.arc thread=1 sequence=2
unable to find archived log
archived log thread=1 sequence=3
RMAN-00571: =====
RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====
RMAN-00571: =====
RMAN-03002: failure of recover command at 05/20/2025 00:49:10
RMAN-06054: media recovery requesting unknown archived log for thread 1 with sequence 3 and starting SCN
of 4917038

```

```
RMAN> run{  
2> set until scn=4917038;  
3> restore database;  
4> recover database;  
5> alter database open resetlogs;  
6> }
```

executing command: SET until clause

Starting restore at 20-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/ORCL/system01.dbf

channel ORA_DISK_1: restoring datafile 00002 to /u01/app/oracle/oradata/ORCL/oetbs01.dbf

channel ORA_DISK_1: restoring datafile 00003 to /u01/app/oracle/oradata/ORCL/sysaux01.dbf

channel ORA_DISK_1: restoring datafile 00004 to /u01/app/oracle/oradata/ORCL/undotbs01.dbf

channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL/shtbs01.dbf

channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: restoring datafile 00008 to /u01/app/oracle/oradata/ORCL/oetbs02.dbf

channel ORA_DISK_1: reading from backup piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T003527_n2pmy0wh_.bkp

channel ORA_DISK_1: piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T003527_n2pmy0wh_.
bkp tag=TAG20250520T003527

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:07

Finished restore at 20-MAY-25

Starting recover at 20-MAY-25

using channel ORA_DISK_1

starting media recovery

archived log for thread 1 with sequence 2 is already on disk as file /home/oracle/arch1/arch_1_2_1201566881.arc

archived log file name=/home/oracle/arch1/arch_1_2_1201566881.arc thread=1 sequence=2

media recovery complete, elapsed time: 00:00:00

Finished recover at 20-MAY-25

Statement processed

new incarnation of database registered in recovery catalog
starting full resync of recovery catalog
full resync complete

```
RMAN> RUN { EXECUTE SCRIPT full_backup; }
```

```
RMAN> list backup;
```

```
RMAN> list archivelog all;
```

```
RMAN> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

<<시나리오 15>> 시간을 기준으로 불완전 복구

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

```
SYS@orcl> create tablespace pmtbs datafile '/u01/app/oracle/oradata/ORCL/pmtbs01.dbf' size 10m;
Tablespace created.
```

```
-- Alert log
```

```
2025-05-20T01:00:27.486903+09:00
```

```
create tablespace pmtbs datafile '/u01/app/oracle/oradata/ORCL/pmtbs01.dbf' size 10m
```

```
Completed: create tablespace pmtbs datafile '/u01/app/oracle/oradata/ORCL/pmtbs01.dbf' size 10m
```

```
---
```

```
SYS@orcl> create tablespace pmtbs datafile '/u01/app/oracle/oradata/ORCL/pmtbs01.dbf' size 10m;
```

```
Tablespace created.
```

```
SYS@orcl> create table hr.dept2 tablespace pmtbs as select * from hr.departments;
```

```
Table created.
```

```
SYS@orcl> select count(*) from hr.dept2;
```

```
COUNT(*)
```

```
-----
```

```
27
```

```
SYS@orcl> select f.tablespace_name, f.file_name, count(*) from dba_extents e,
dba_data_files f where e.file_id = f.file_id and e.segment_name = 'DEPT2'
and e.owner = 'HR' group by f.tablespace_name, f.file_name;
```

TABLESPACE_NAME	FILE_NAME	COUNT(*)
PMTBS	/u01/app/oracle/oradata/ORCL/pmtbs01.dbf	1

```
RMAN> report need backup;
```

```
starting full resync of recovery catalog
```

```
full resync complete
```

```
RMAN retention policy will be applied to the command
```

```
RMAN retention policy is set to redundancy 1
```

Report of files with less than 1 redundant backups

File #bkps Name

```
-----
9      0      /u01/app/oracle/oradata/ORCL/pmtbs01.dbf
```

RMAN> backup tablespace pmtbs;

Starting backup at 20-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting full datafile backup set

channel ORA_DISK_1: specifying datafile(s) in backup set

input datafile file number=00009 name=/u01/app/oracle/oradata/ORCL/pmtbs01.dbf

channel ORA_DISK_1: starting piece 1 at 20-MAY-25

channel ORA_DISK_1: finished piece 1 at 20-MAY-25

piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T010437_n2ponoo5_.

bkp tag=TAG20250520T010437 comment=NONE

channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01

Finished backup at 20-MAY-25

Starting Control File and SPFILE Autobackup at 20-MAY-25

piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201568679_n2ponql2_.bkp

comment=NONE

Finished Control File and SPFILE Autobackup at 20-MAY-25

RMAN> list backup;

RMAN> list backup of tablespace pmtbs;

List of Backup Sets

=====

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time
--------	------	---------	-------------	--------------	-----------------

3059	Full	1.09M	DISK	00:00:00	20-MAY-25
------	------	-------	------	----------	-----------

BP Key: 3062 Status: AVAILABLE Compressed: NO Tag: TAG20250520T010437

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T010437_n2ponoo5_.bkp

List of Datafiles in backup set 3059

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	----	------	---------	----------	-------------	--------	------

9		Full	4922207	20-MAY-25		NO	/u01/app/oracle/oradata/ORCL/pmtbs01.dbf
---	--	------	---------	-----------	--	----	--

- 장애 발생

SYS@orcl> drop tablespace pmtbs including contents and datafiles;
Tablespace dropped.

RMAN> report schema;
starting full resync of recovery catalog
full resync complete
Report of database schema for database with db_unique_name ORCL
List of Permanent Datafiles

```
=====
```

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	620	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs02.dbf

List of Temporary Files

```
=====
```

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32	/u01/app/oracle/oradata/ORCL/temp01.dbf

* 필요한 정보

- pmtbs t/s 삭제 시간: 2025-05-20T01:06:23.878342+09:00
- CF : /home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201568679_n2ponql2_.bkp

RMAN> shutdown immediate
RMAN> startup nomount;

RMAN> alter database mount;

Statement processed

RMAN> report schema;

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	620	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs02.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32	/u01/app/oracle/oradata/ORCL/temp01.dbf

```

RMAN> alter session set nls_date_format='yyyy-mm-dd hh24:mi:ss';

```

```

Statement processed

```

```

RMAN> select sysdate from dual;

```

```

SYSDATE

```

```

2025-05-20 01:19:18

```

```

RMAN> run {

```

```

2> set until time '2025-05-20 01:06:22';

```

```

3> restore database;

```

```

4> recover database;

```

```

5> alter database open resetlogs;

```

```

6> }

```

```

-- 완료후 백업받기

```

```

RMAN> RUN { EXECUTE SCRIPT full_backup; }

```

```

RMAN> list backup;

```

```

RMAN> list archivelog all;

```

```

RMAN> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;

```